Chapter 37

Wave Optics (quang học sóng)

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Wave Optics

Wave optics is a study concerned with phenomena that cannot be adequately explained by geometric (ray) optics.

- Sometimes called *physical optics*

These phenomena include:

- Interference (giao thoa)
- Diffraction (nhiễu xạ)
- Polarization (phân cực)

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Interference (giao thoa)

In *constructive interference (giao thoa tăng cường)* the amplitude of the resultant wave is greater than that of either individual wave.

In *destructive interference (giao thoa triệt tiêu)* the amplitude of the resultant wave is less than that of either individual wave.

All interference associated with light waves arises when the electromagnetic fields that constitute the individual waves combine.

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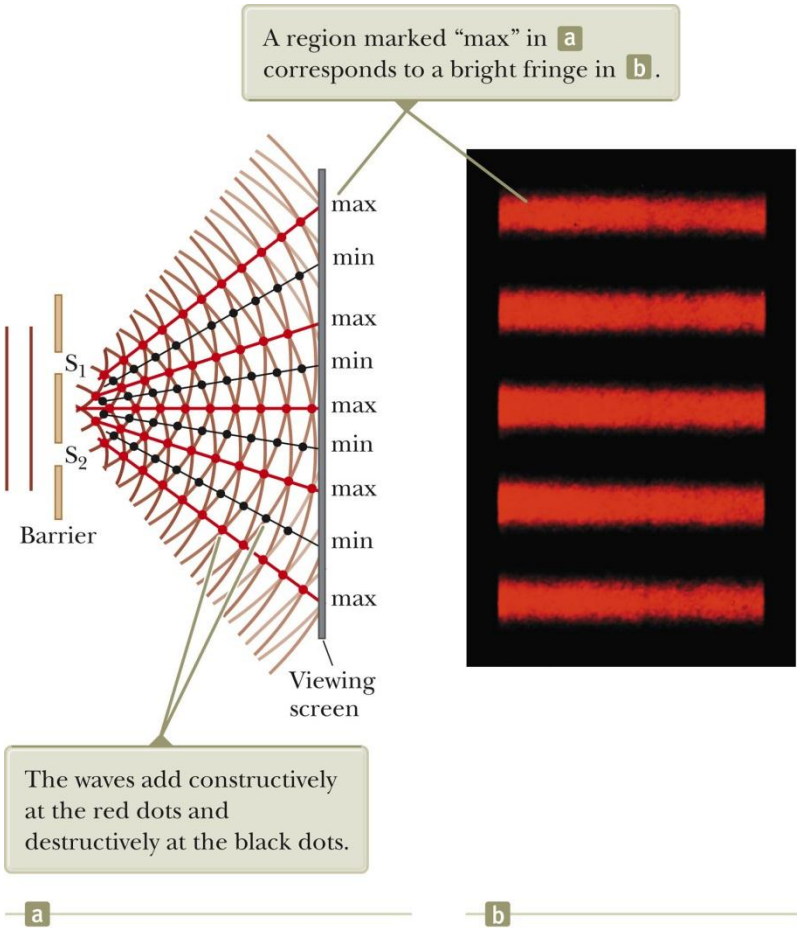
Young's Double-Slit Experiment: Schematic

Thomas Young first demonstrated interference in light waves from two sources in 1801.

The narrow slits S_1 and S_2 act as sources of waves.

The waves emerging from the slits (khe) originate from the same wave front and therefore are always in phase.



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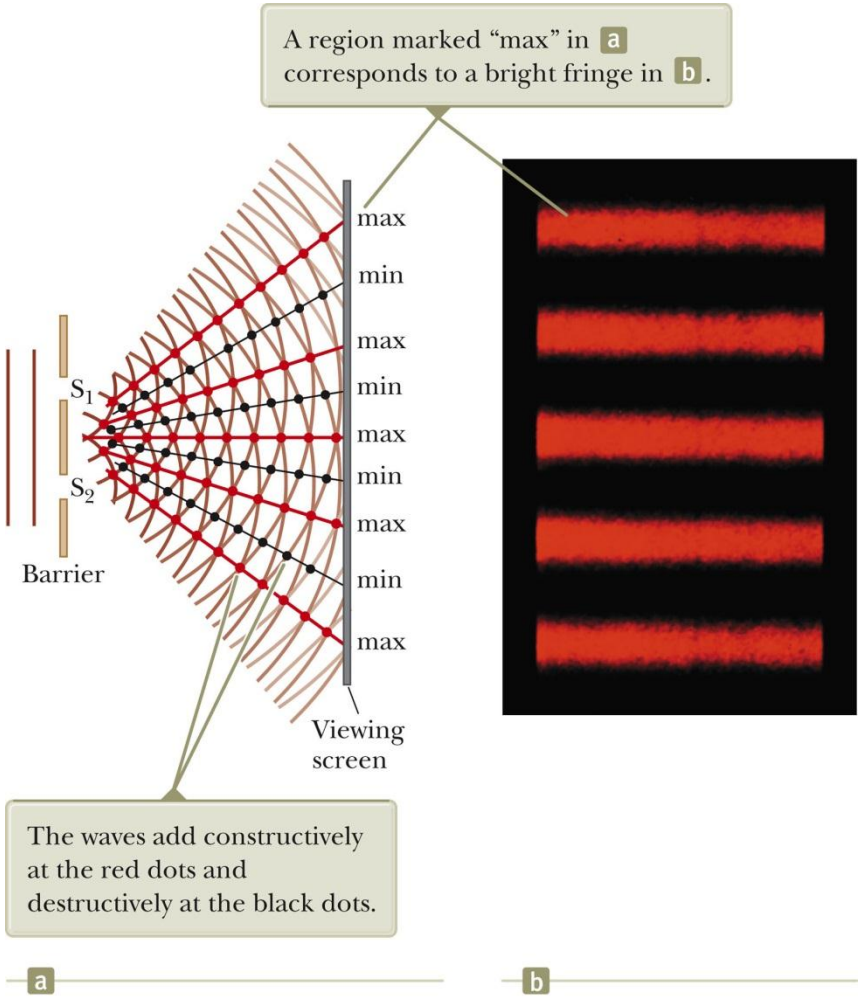
Resulting Interference Pattern

The light from the two slits forms a visible pattern on a screen.

The pattern consists of a series of bright and dark parallel bands called *fringes (vân).*

Constructive interference occurs where a bright fringe (vân sáng) occurs.

Destructive interference results in a dark fringe (vân tối).



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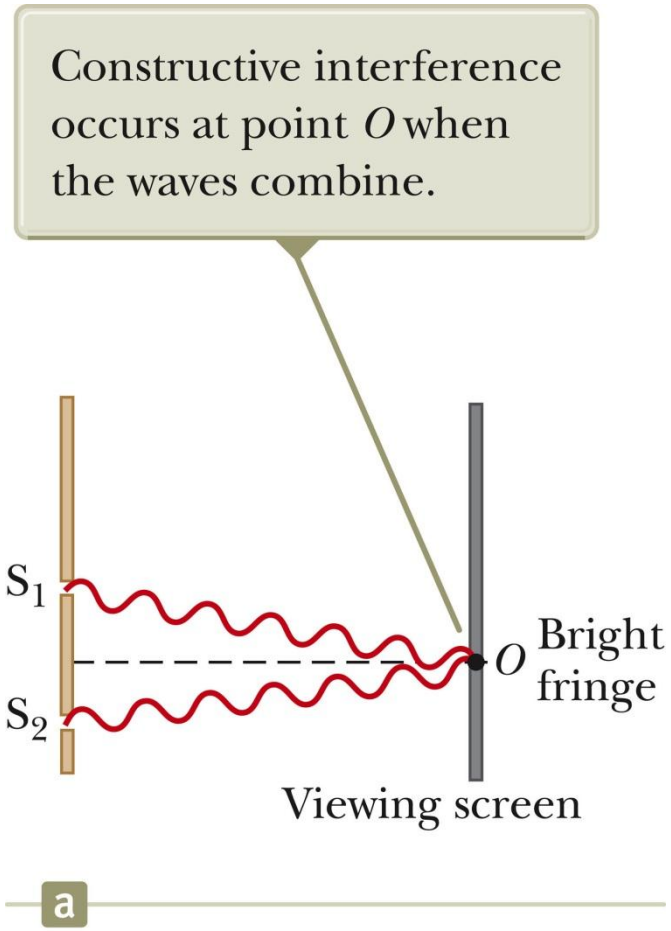
Interference Patterns

Constructive interference occurs at point O.

The two waves travel the same distance.

- Therefore, they arrive in phase

As a result, constructive interference occurs at this point and a bright fringe is observed.



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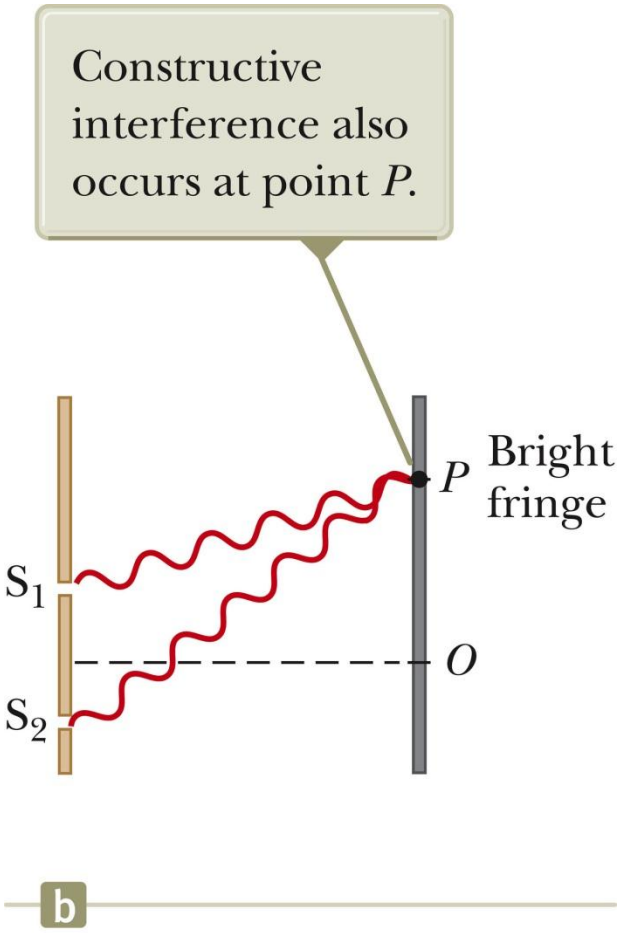
Interference Patterns, 2

The lower wave has to travel farther than the upper wave to reach point *P.*

The lower wave travels one wavelength farther.

- Therefore, the waves arrive in phase

A second bright fringe occurs at this position.



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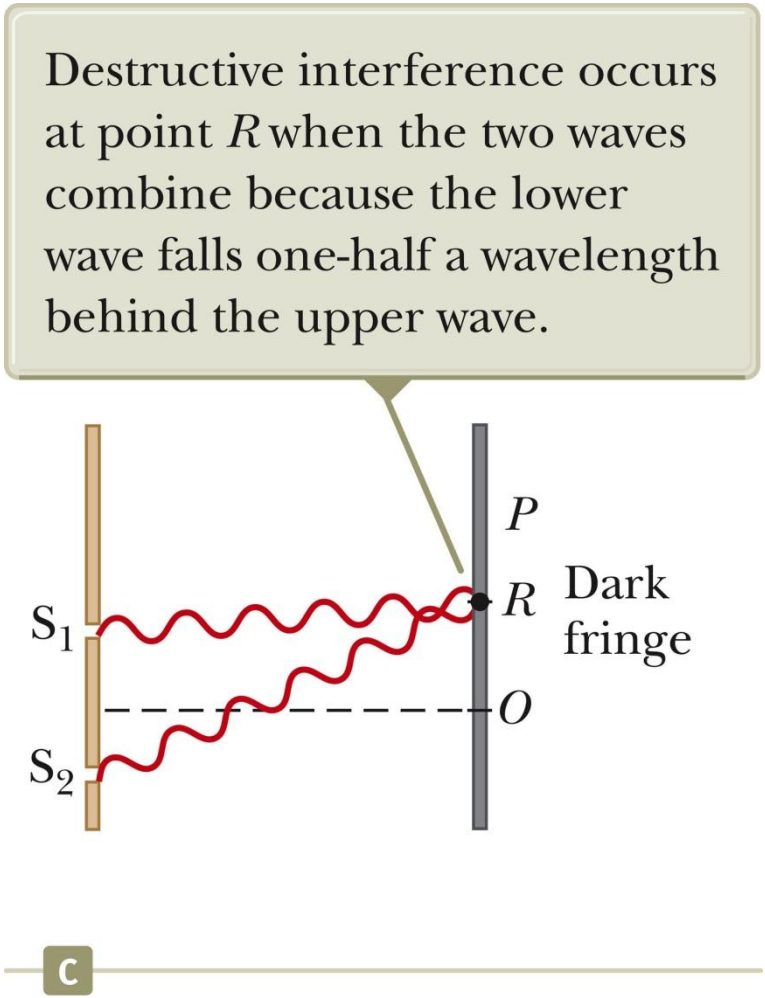
Interference Patterns, 3

The upper wave travels one-half of a wavelength farther than the lower wave to reach point *R.*

The trough of the upper wave overlaps the crest of the lower wave.

This is destructive interference.

- A dark fringe occurs.



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Conditions for Interference

To observe interference in light waves, the following two conditions must be met:

- The sources must be **coherent (kết hợp).**
  - They must maintain a constant phase with respect to each other.
- The sources should be **monochromatic (đơn sắc).**
  - Monochromatic means they have a single wavelength.

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Producing Coherent Sources

Light from a monochromatic source is used to illuminate a barrier.

The barrier contains two small openings.

- The openings are usually in the shape of slits.

The light emerging from the two slits is coherent since a single source produces the original light beam.

This is a commonly used method.

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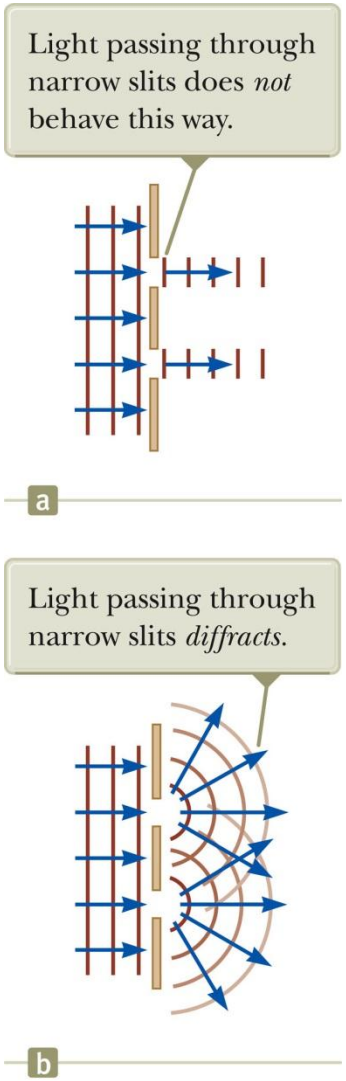
Diffraction (nhiễu xạ)

If the light traveled in a straight line after passing through the slits, no interference pattern would be observed.

From Huygens's principle we know the waves spread out from the slits.

This divergence of light from its initial line of travel is called **diffraction.**



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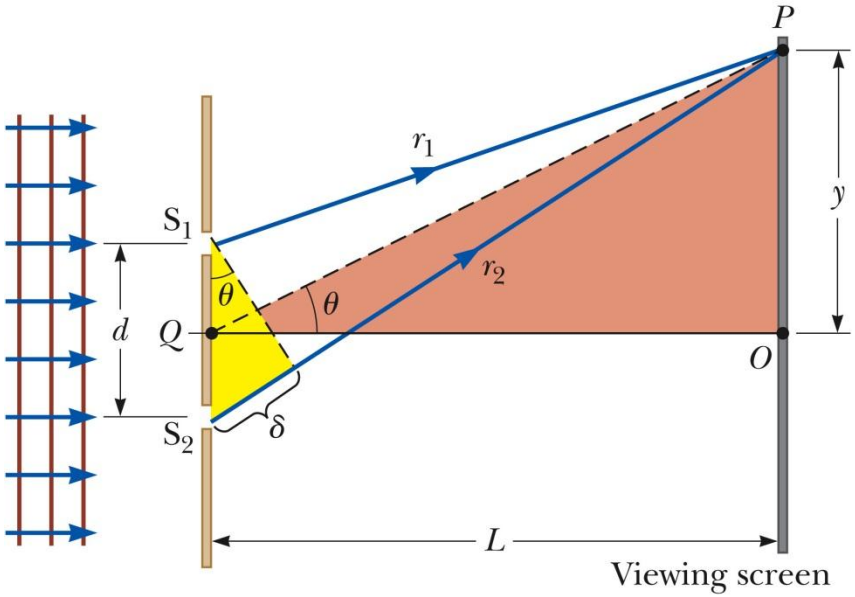
Young's Double-Slit Experiment: Geometry

The path difference, δ, is found from geometry.

$\delta = r_2 - r_1 = d \sin \theta$

- This assumes the paths are parallel.
- Not exactly true, but a very good approximation if L is much greater than d



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Interference Equations

For a bright fringe produced by constructive interference, the path difference must be either zero or some integer multiple of the wavelength.

$\delta = d \sin \theta_{bright} = m\lambda$

- m = 0, ±1, ±2, …
- m is called the order number
  - When m = 0, it is the zeroth-order maximum
  - When m = ±1, it is called the first-order maximum

When destructive interference occurs, a dark fringe is observed.

This needs a path difference of an odd half wavelength.

$\delta = d \sin \theta_{dark} = (m + ½)\lambda$

- m = 0, ±1, ±2, …

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Interference Equations, cont.

The positions of the fringes can be measured vertically from the zeroth-order maximum.

Using the large triangle in fig. 37.5,

- $y_{bright} = L \tan \theta_{bright}$
- $y_{dark} = L \tan \theta_{dark}$

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Interference Equations, final

Assumptions in a Young's Double Slit Experiment:

- L >> d
- d >> λ

Approximation:

- θ is small and therefore the small angle approximation tan θ ~ sin θ can be used

y = L tan θ ≈ L sin θ

For small angles,

$$y_{\text{bright}} = L\frac{m\lambda}{d} \text{ and } y_{\text{dark}} = L\frac{(m+1/2)\lambda}{d}$$

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Uses for Young's Double-Slit Experiment

Young's double-slit experiment provides a method for measuring wavelength of the light.

This experiment gave the wave model of light a great deal of credibility.

- It was inconceivable that particles of light could cancel each other in a way that would explain the dark fringes.

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Intensity Distribution: Double-Slit Interference Pattern

The bright fringes in the interference pattern do not have sharp edges.

- The equations developed give the location of only the centers of the bright and dark fringes.

We can calculate the distribution of light intensity associated with the double-slit interference pattern.

Assumptions:

- The two slits represent coherent sources of sinusoidal waves.
- The waves from the slits have the same angular frequency, ω.
- The waves have a constant phase difference, φ.

The total magnitude of the electric field at any point on the screen is the superposition of the two waves.

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Intensity Distribution, Electric Fields

The magnitude of each wave at point *P* on a screen can be found.

- $E_1 = E_o \sin \omega t$
- $E_2 = E_o \sin (\omega t + \varphi)$
  - Both waves have the same amplitude, E_o.

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Intensity Distribution, Phase Relationships

The phase difference between the two waves at P depends on their path difference.

- $\delta = r_2 - r_1 = d \sin \theta$

A path difference of λ (for constructive interference) corresponds to a phase difference of 2π rad.

A path difference of δ is the same fraction of λ as the phase difference φ is of 2π.

This gives

$$\varphi = \frac{2\pi}{\lambda}\delta = \frac{2\pi}{\lambda} d \sin \theta$$

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Intensity Distribution, Resultant Field

The magnitude of the resultant electric field comes from the superposition principle.

- $E_P = E_1 + E_2 = E_o[\sin \omega t + \sin (\omega t + \varphi)]$

This can also be expressed as

$$E_P = 2E_o \cos\left(\frac{\varphi}{2}\right)\sin\left(\omega t + \frac{\varphi}{2}\right)$$

- E_P has the same frequency as the light at the slits.
- The magnitude of the field is multiplied by the factor $2 \cos (\varphi / 2)$.

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Intensity Distribution, Equation

The expression for the intensity comes from the fact that the *intensity of a wave is proportional to the square of the resultant electric field magnitude at that point.*

The intensity therefore is

$$I = I_{\max} \cos^2\left(\frac{\pi d \sin\theta}{\lambda}\right) \approx I_{\max} \cos^2\left(\frac{\pi d}{\lambda L} y\right) \quad (\textit{small angles})$$

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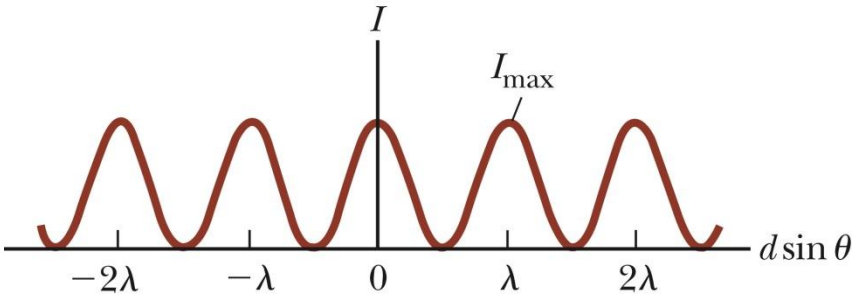
Light Intensity, Graph

The interference pattern consists of equally spaced fringes of equal intensity



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Multiple Slits, Graph

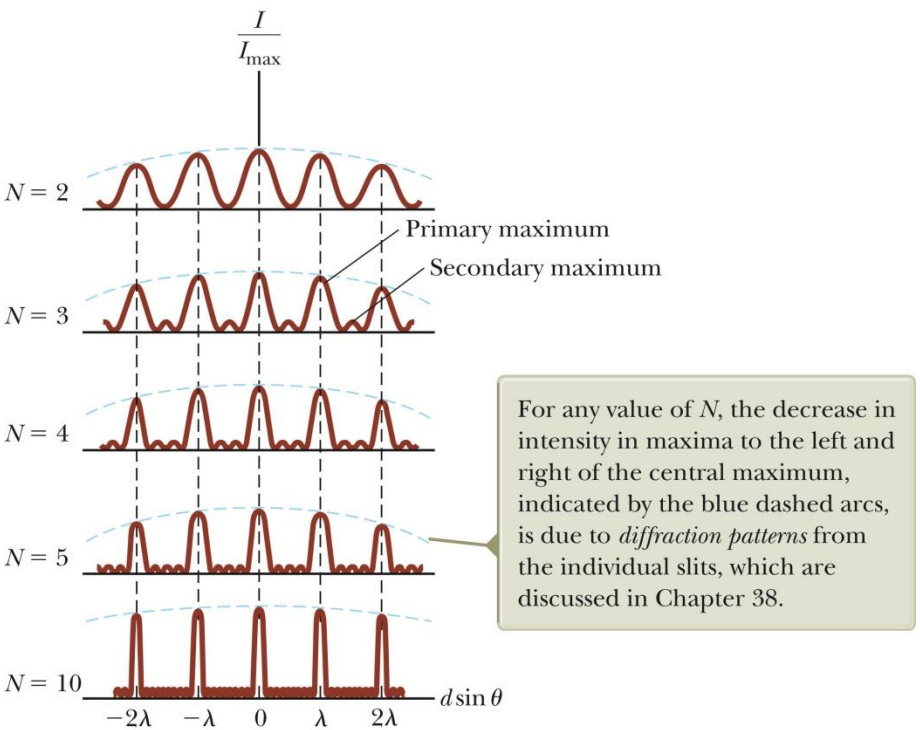
With more than two slits, the pattern contains primary and secondary maxima.

For N slits, the intensity of the primary maxima is N^2 times greater than that due to a single slit.

As the number of slits increases, the primary maxima increase in intensity and become narrower.

- The secondary maxima decrease in intensity relative to the primary maxima.

The number of secondary maxima is N – 2, where N is the number of slits.



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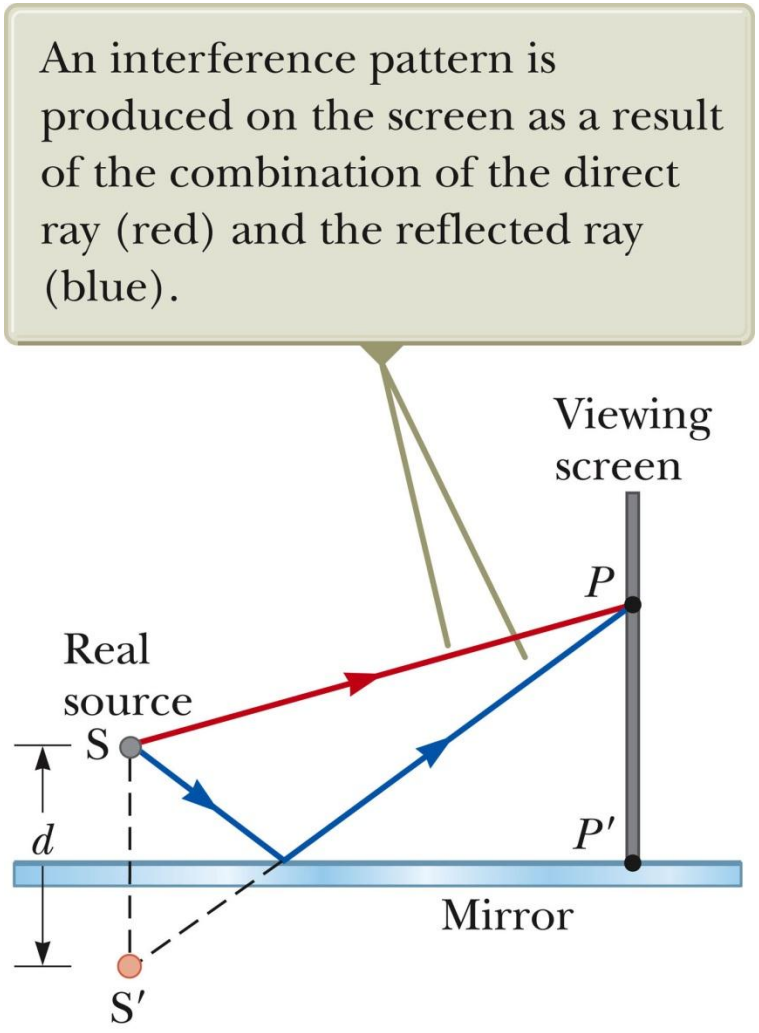
Lloyd's Mirror

An arrangement for producing an interference pattern with a single light source.

Waves reach point *P* either by a direct path or by reflection.

The reflected ray can be treated as a ray from the source S' behind the mirror.



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Interference Pattern from a Lloyd's Mirror

This arrangement can be thought of as a double-slit source with the distance between points S and S' comparable to length *d.*

An interference pattern is formed.

The positions of the dark and bright fringes are reversed relative to the pattern of two real sources.

This is because there is a 180° phase change produced by the reflection.

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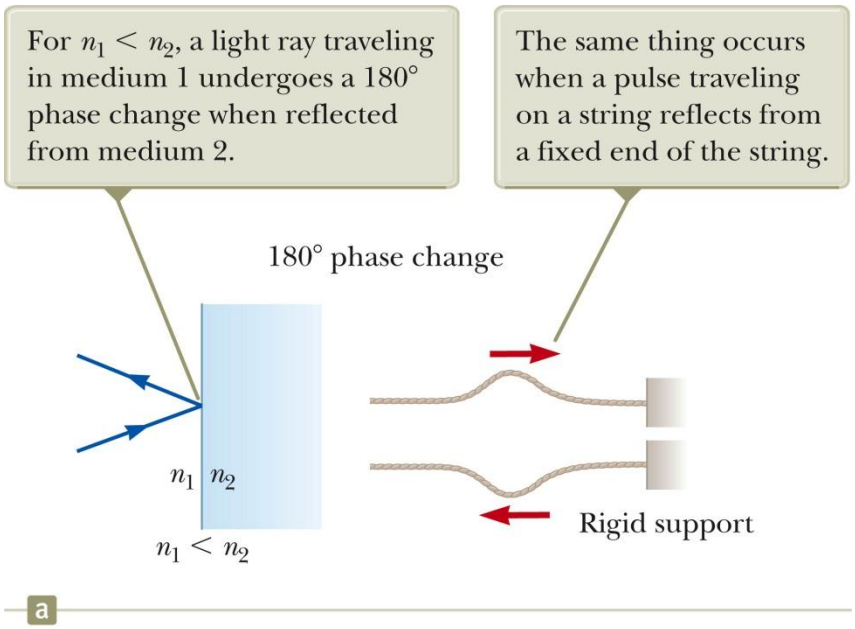
Phase Changes Due To Reflection

An electromagnetic wave undergoes a phase change of 180° upon reflection from a medium of higher index of refraction than the one in which it was traveling.

- Analogous to a pulse on a string reflected from a rigid support



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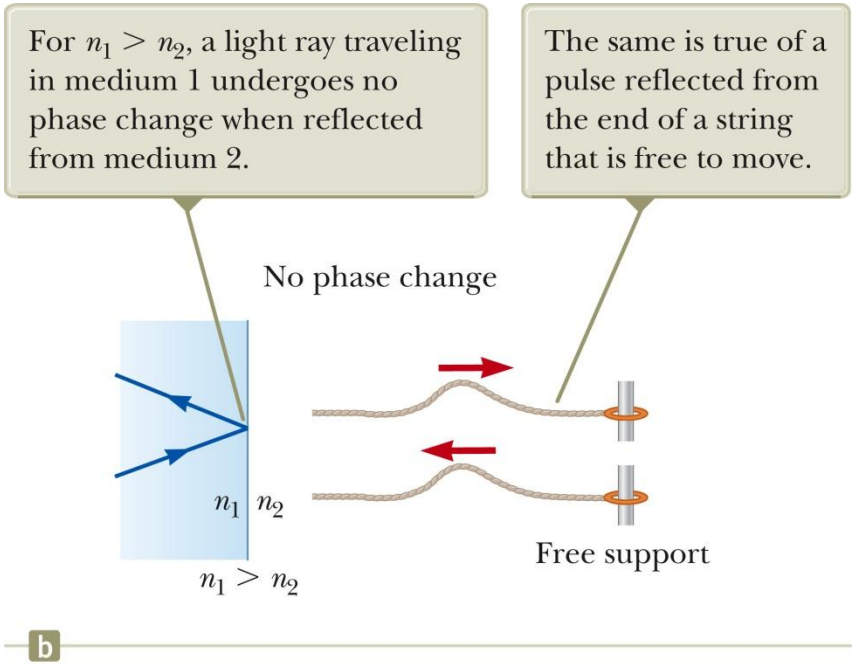
Phase Changes Due To Reflection, cont.

There is no phase change when the wave is reflected from a boundary leading to a medium of lower index of refraction.

- Analogous to a pulse on a string reflecting from a free support



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Interference in Thin Films

Interference effects are commonly observed in thin films.

- Examples include soap bubbles and oil on water

The various colors observed when white light is incident on such films result from the interference of waves reflected from the two surfaces of the film.

Facts to remember:

- An electromagnetic wave traveling from a medium of index of refraction n_1 toward a medium of index of refraction n_2 undergoes a 180° phase change on reflection when $n_2 > n_1$.
  - There is no phase change in the reflected wave if $n_2 < n_1$.
- The wavelength of light λ_n in a medium with index of refraction n is $\lambda_n = \lambda/n$ where λ is the wavelength of light in vacuum.

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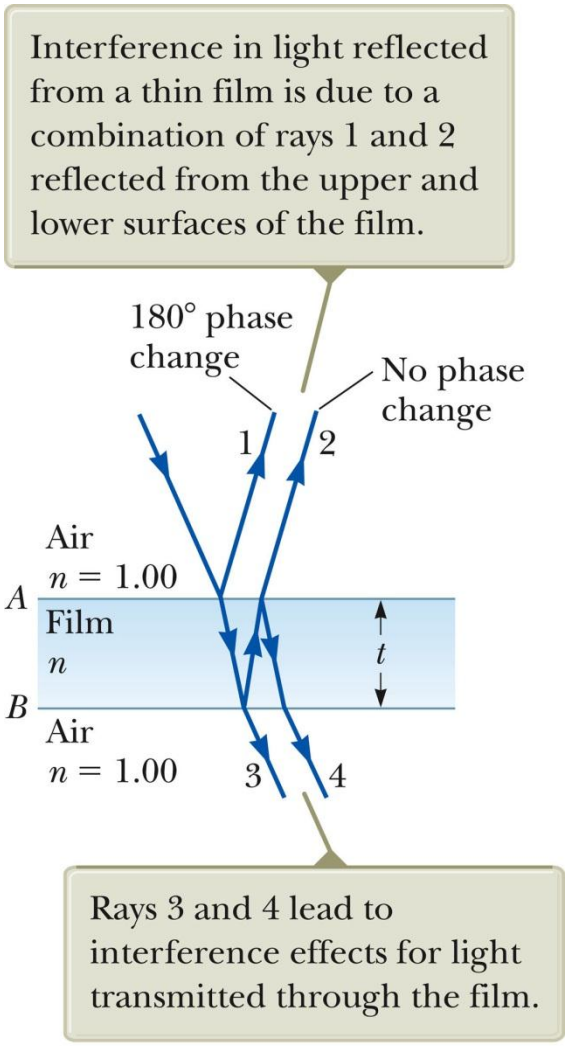
Interference in Thin Films, 2

Assume the light rays are traveling in air nearly normal to the two surfaces of the film.

Ray 1 undergoes a phase change of 180° with respect to the incident ray.

Ray 2, which is reflected from the lower surface, undergoes no phase change with respect to the incident wave.



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Interference in Thin Films, 3

Ray 2 also travels an additional distance of 2t before the waves recombine.

For constructive interference

- $2nt = (m + ½)\lambda$ $(m = 0, 1, 2 \ldots)$
  - This takes into account both the difference in optical path length for the two rays and the 180° phase change.

For destructive interference

- $2nt = m\lambda$ $(m = 0, 1, 2 \ldots)$

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Interference in Thin Films, 4

Two factors influence interference

- Possible phase reversals on reflection
- Differences in travel distance

The conditions are valid if the medium above the top surface is the same as the medium below the bottom surface.

- If there are different media, these conditions are valid as long as the index of refraction for both is less than n.

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Interference in Thin Films, 5

If the thin film is between two different media, one of lower index than the film and one of higher index, the conditions for constructive and destructive interference are reversed.

With different materials on either side of the film, you may have a situation in which there is a 180° phase change at both surfaces or at neither surface.

- Be sure to check both the path length and the phase change.

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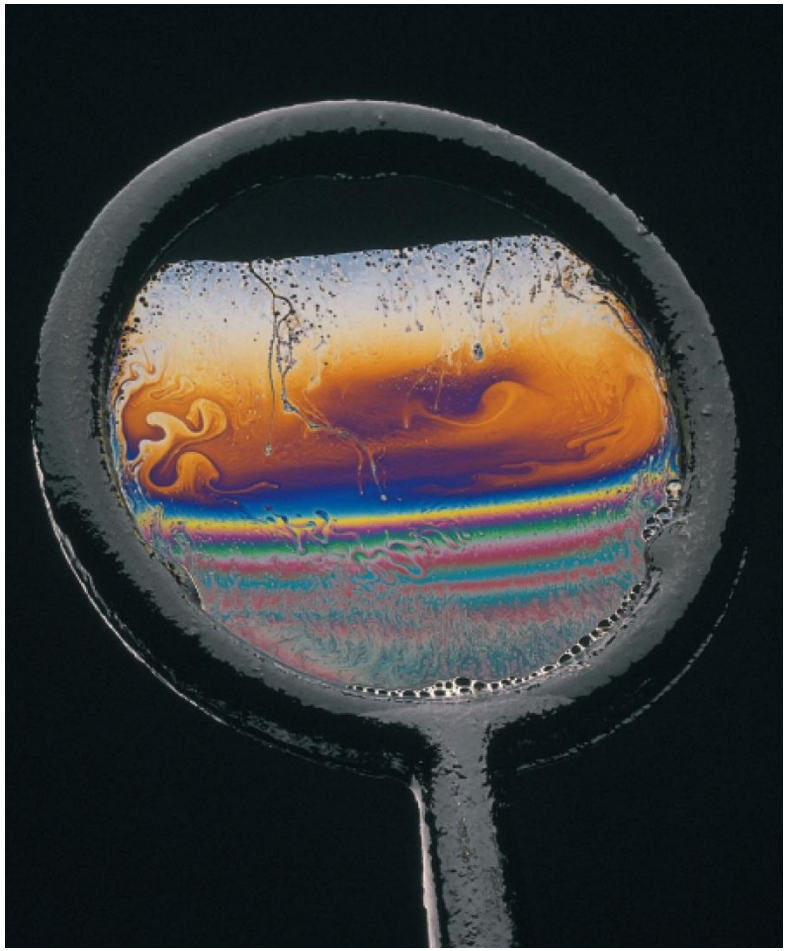
Interference in Thin Film, Soap Bubble Example



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Newton's Rings

Another method for viewing interference is to place a plano-convex lens on top of a flat glass surface.

The air film between the glass surfaces varies in thickness from zero at the point of contact to some thickness t.

A pattern of light and dark rings is observed.

- These rings are called Newton's rings.
- The particle model of light could not explain the origin of the rings.

Newton's rings can be used to test optical lenses.

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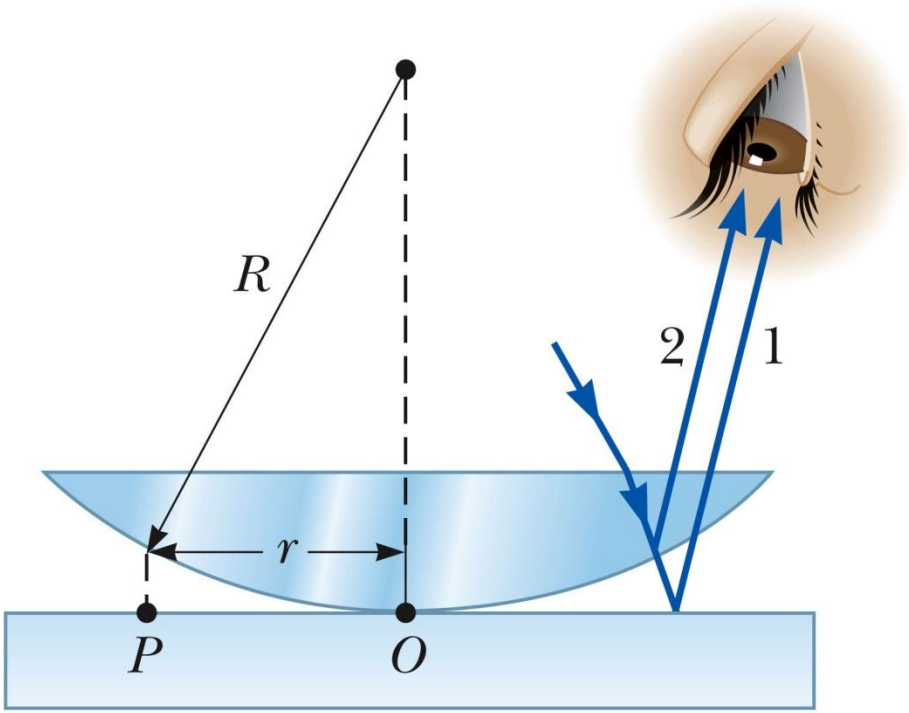
Newton's Rings, Set-Up and Pattern



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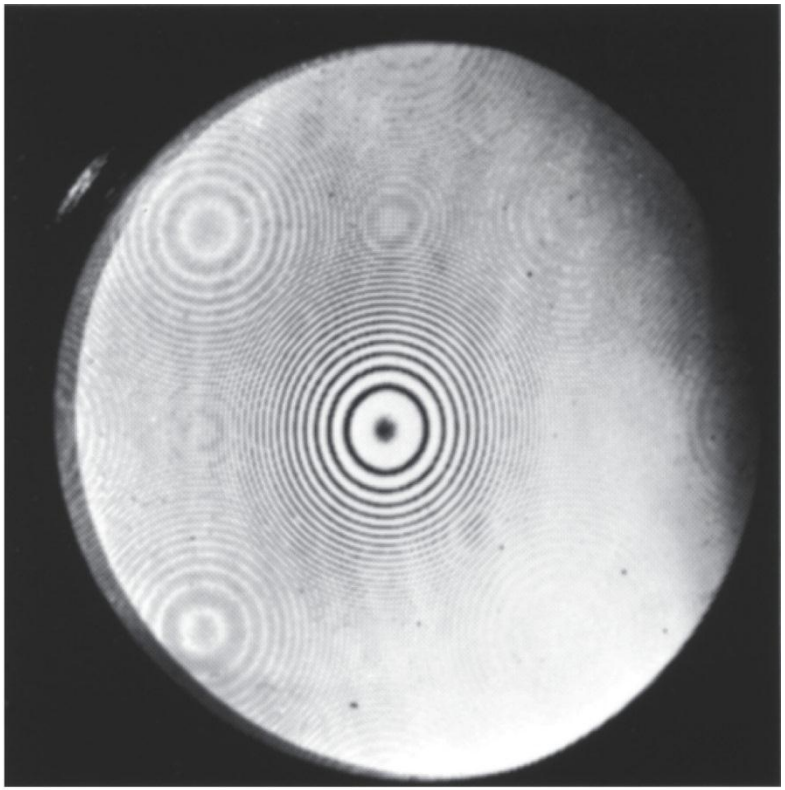

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Problem Solving Strategy with Thin Films, 1

Conceptualize

- Identify the light source.
- Identify the location of the observer.

Categorize

- Be sure the techniques for thin-film interference are appropriate.
- Identify the thin film causing the interference.

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Problem Solving with Thin Films, cont.

Analyze

- The type of interference – constructive or destructive – that occurs is determined by the phase relationship between the upper and lower surfaces.
- Phase differences have two causes.
  - differences in the distances traveled
  - phase changes occurring on reflection
- Both causes must be considered when determining constructive or destructive interference.
- Use the indices of refraction of the materials to determine the correct equations.

Finalize

- Be sure your results make sense physically.
- Be sure they are of an appropriate size.

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Michelson Interferometer

The interferometer was invented by an American physicist, A. A. Michelson.

The interferometer splits light into two parts and then recombines the parts to form an interference pattern.

The device can be used to measure wavelengths or other lengths with great precision.

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Michelson Interferometer, Schematic

A ray of light is split into two rays by the mirror M_o.

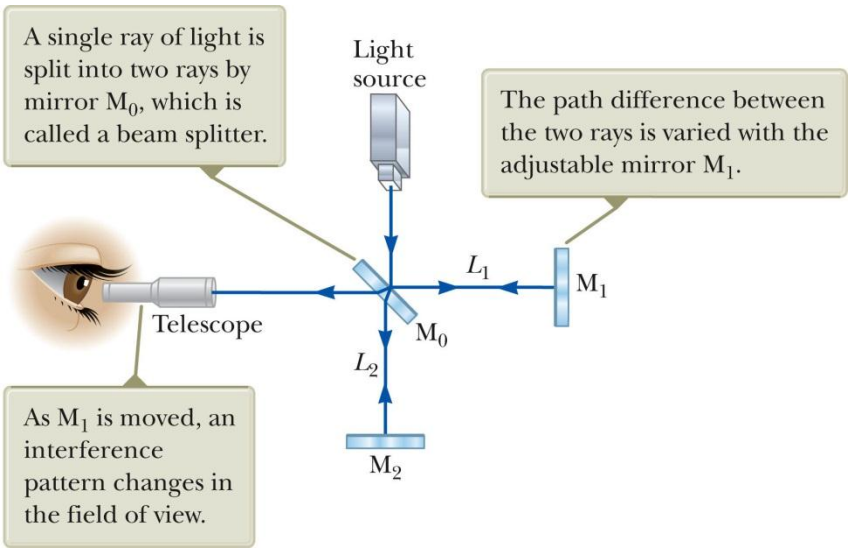
- The mirror is at 45° to the incident beam.
  - The mirror is called a *beam splitter.*
  - It transmits half the light and reflects the rest.

The reflected ray goes toward mirror M_1.

The transmitted ray goes toward mirror M_2.

The two rays travel separate paths L_1 and L_2.

After reflecting from M_1 and M_2, the rays eventually recombine at M_o and form an interference pattern.



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Michelson Interferometer – Operation

The interference condition for the two rays is determined by their path length difference.

M_1 is moveable.

As it moves, the fringe pattern collapses or expands, depending on the direction M_1 is moved.

The fringe pattern shifts by one-half fringe each time M_1 is moved a distance $\lambda/4$.

The wavelength of the light is then measured by counting the number of fringe shifts for a given displacement of M_1.

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Michelson Interferometer – Applications

The Michelson interferometer was used to disprove the idea that the Earth moves through an ether.

Modern applications include:

- Fourier Transform Infrared Spectroscopy (FTIR)
- Laser Interferometer Gravitational-Wave Observatory (LIGO)

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Fourier Transform Infrared Spectroscopy

This is used to create a high-resolution spectrum in a very short time interval.

The result is a complex set of data relating light intensity as a function of mirror position.

- This is called an interferogram.

The interferogram can be analyzed by a computer to provide all of the wavelength components.

- This process is called a Fourier transform.

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Laser Interferometer Gravitational-Wave Observatory

General relativity predicts the existence of gravitational waves.

In Einstein's theory, gravity is equivalent to a distortion of space.

- These distortions can then propagate through space.

The LIGO apparatus is designed to detect the distortion produced by a disturbance that passes near the Earth.

The interferometer uses laser beams with an effective path length of several kilometers.

At the end of an arm of the interferometer, a mirror is mounted on a massive pendulum.

When a gravitational wave passes, the pendulum moves, and the interference pattern due to the laser beams from the two arms changes.

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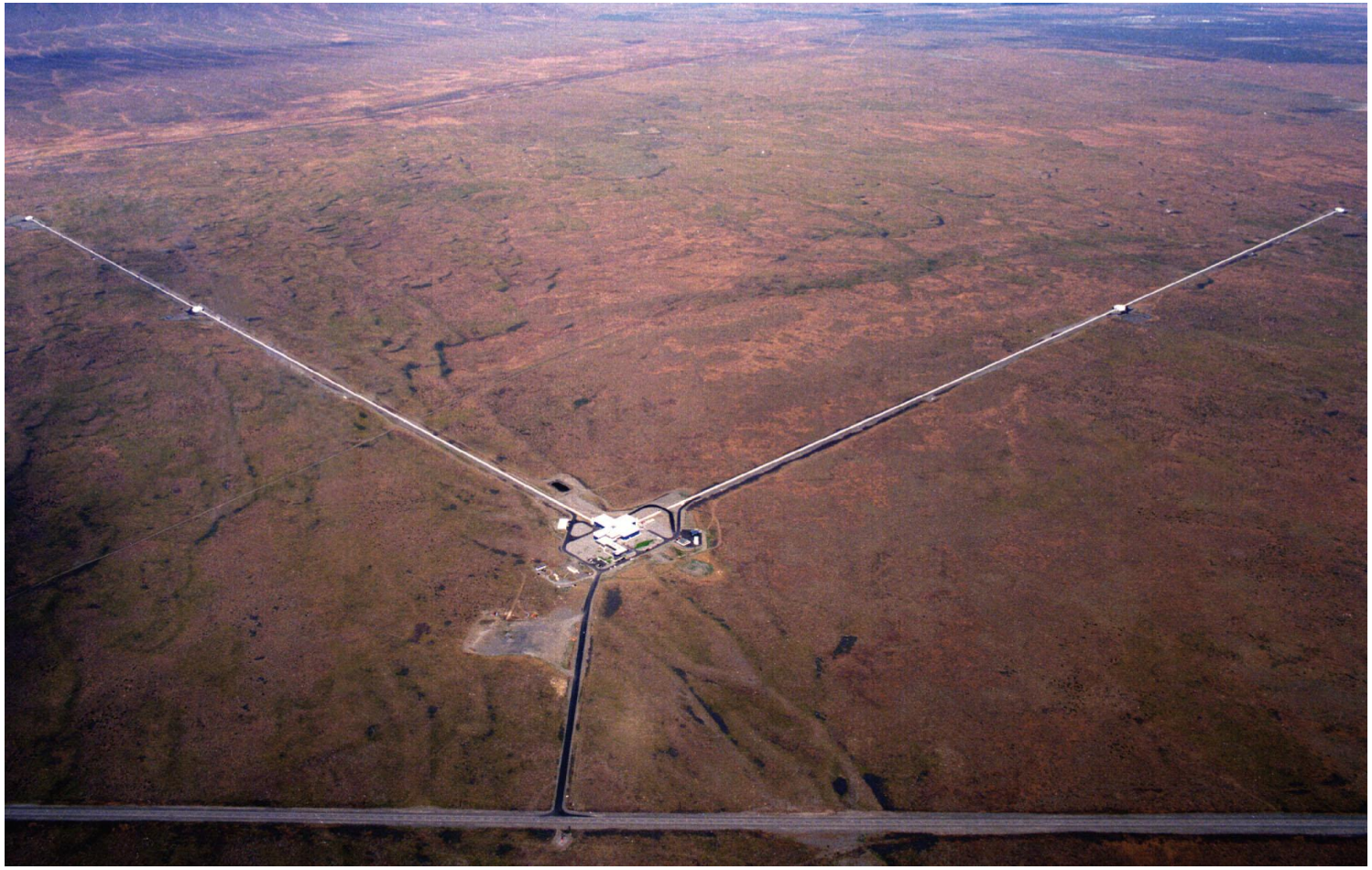
LIGO in Richland, Washington



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